

Real-time 52.8 Tb/s (26.4Tb/s/core) Unrepeated Transmission over 303.4km of Uncoupled 2-core MCF

B. Zhu^{(1)*}, M. Stegmaier⁽²⁾, T. Geisler⁽²⁾, P. I. Borel⁽²⁾, B. Palsdottir⁽²⁾, P. Jenneve⁽³⁾, H. Zhang⁽⁴⁾

⁽¹⁾ Lightera Labs, Somerset, NJ USA, benyuan.zhu@lightera.com

⁽²⁾ Lightera Broendby, Denmark

⁽³⁾ Cisco System, Maynard MA USA

⁽⁴⁾ Cisco System, Holmdel NJ USA

Abstract We demonstrate real-time unrepeated transmission of 52.8 Tb/s over 303.4 km uncoupled 2-core MCF using commercial pluggable modules and ultra-low loss SCUBA2X fibre (0.147 dB/km). We achieve 26.4 Tb/s per core (33×800Gb/s/core) and show no measurable inter-core crosstalk penalty in both unidirectional and bidirectional operations. ©2026 The Author(s)

Introduction

To meet escalating capacity demands, the submarine industry is adopting space-division multiplexing (SDM) via multi-core fibre (MCF) designed to fit existing cable dimensions. Uncoupled two-core MCF has emerged as a leading candidate for immediate deployment as its 125 μm cladding avoids the need for extensive cable redesign [1]. Its negligible inter-core crosstalk (XT) enables the use of standard transponders and avoids complex MIMO DSP, allowing operators to scale capacity using proven, commercially available hardware.

The performance of SDM systems is primarily governed by two critical factors: fiber attenuation and inter-core XT. In long-haul repeated systems, bidirectional configurations have been effectively employed to mitigate XT impacts [2-5]. However, recent studies on unrepeated systems suggest that bidirectional transmission may not offer the same XT reduction advantages, particularly in unrepeated long-distance links [6-7]. Unrepeated systems maximize capacity over long distances by eliminating in-line active elements, thus they reduce system complexity and cost, offering an economical solution for undersea links and remote terrestrial routes in hostile environments. Recent significant progresses have been demonstrated in single-core fibre (SCF), including 25.6 Tb/s real-time transmission over 320 km [8], 41.3 Tb/s C-band transmission over 291 km [9], and a record 66.8

Tb/s real-time C+L band transmission over 301 km [10]. In the realm of MCF, 372.8 Tb/s (total) has been achieved over 213 km using 4-core MCF [11]; real-time 25.2 Tb/s/core unrepeated transmission was demonstrated over 256 km of 2-core MCF [6]. The unrepeated distances have been extended to 359 km, but capacity was significantly reduced to 8.4 Tb/s [7].

In this work, we demonstrate a record real-time unrepeated transmission of 26.4 Tb/s per core (52.8 Tb/s total) over a 303.4 km link. This was achieved using commercially pluggable modules and SCUBA 2X 2-core MCF featuring an ultra-low average attenuation of 0.147 dB/km [12]. We provide the measurement data of co-propagation and counter-propagation inter-core XT across various lengths, exhibiting exceptional low co-propagation crosstalk (CO-XT) with average -59dB/100km. Our results confirm that even in a high-power DWDM transmission configuration using backward-pumped distributed Raman amplification (DRA), the inter-core XT induces no measurable penalty in either unidirectional or bidirectional unrepeated transmissions. These findings validate the readiness of SCUBA 2X for high capacity terrestrial and subsea unrepeated branches.

Experimental Set-up

The experimental configuration for the 2-core MCF unrepeated transmission is illustrated in Fig. 1. The transmitter stage consists of a

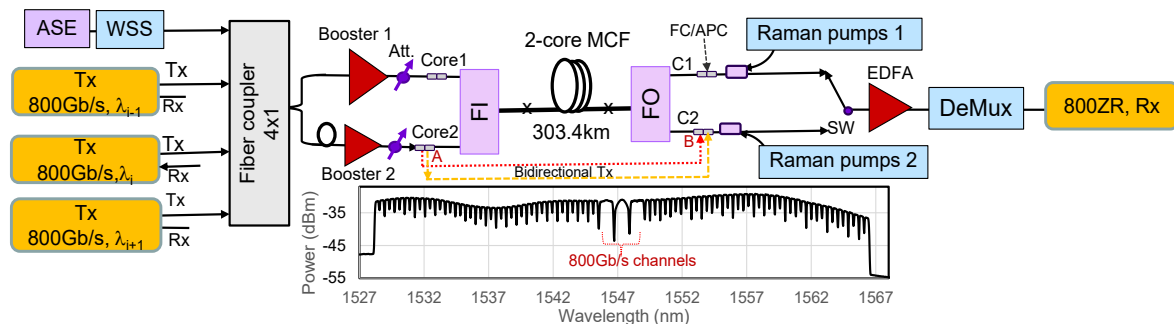


Fig. 1: Experimental set-up for 2-core MCF unrepeated transmission, inset: example of received optical spectra.

measurement path and a loading path. For the measurement path, three real-time pluggable transceivers with adjustable bitrates and modulation formats are used. We choose to operate at 138 GBd with probabilistic constellation shaped (PCS)-16QAM modulation, resulting in 800 Gb/s net data rate per wavelength, considering the FEC overhead and pilot tone etc. We transmit 33 channels within the C-band, ranging from 1528.77 to 1567.13 nm with 150 GHz channel spacing. The wavelengths of our 3 real-time pluggable modules are set to adjacent channels λ_{i-1} , λ_i and λ_{i+1} respectively. The loading path is composed of a broadband ASE source and is notch-filtered by a 50GHz channelized wavelength selective switch (WSS) filter for a bandwidth window of 450 GHz. The combined signals are amplified and then is split over two paths, decorrelated, and sent to both MCF cores via fan-in/fan-out (FIFO). At the receiver end, the DWDM signals are demultiplexed by another WSS, the (bit-error rate) BER and OSNR of the channel at λ_i are measured, then the wavelengths are tuned to the next channels. When assessing the system performance, these three 800Gb/s channels are loaded with equal launch powers with ASE channels and translated across the C-band. Example of a received optical spectra is shown in inset of Fig.1

The transmission link spans 303.4 km and consists of eight spools of uncoupled 2-core MCF. Key fibre parameters at 1550 nm are summarized in Table 1. Inter-core XT was characterized using an ASE source across various distances, including a single spool (22.6 km), three 108.5 km spans, and the full 303.4 km link. The measured average CO-XT and counter-XT values are presented in Table 2, showing exceptional low co-propagation XT with average -59dB/100km. It can be seen that the counter-XT was increased to -48.2dB (higher than CO-XT) at 303.4km full link length, which aligns with the trends reported in [13]. Note that all inter-core XT measurements, including those for the 303.4 km link, were conducted without distributed Raman

Table 1: Key fiber parameters at 1550nm

Core	Loss (dB/km)	A_{eff} (mm ²)	MFD (μm)	PMD (ps/km ^{0.5})	λ_c (nm)
1	0.147	110.1	11.6	0.06	1454
2	0.147	110.5	11.6	0.07	1454

Table 2: Measured CO-XT and Counter-XT from difference length of SCUBA 2X 2-core MCF

MCF length (km)	CO-XT (dB)	Counter-XT (dB)
22.6	-64.8	
108.5	-58.7	<-80
303.4	-53.9	-48.2

amplification (DRA) due to high ASE floor, and the counter-XT for the 22.6 km spool could not be measured due to power meter sensitivity limitations. All reported XT measurements include a FIFO pair, with each FIFO contributing an average XT of -67 dB.

The total link losses for Core 1 and Core 2 were 47.2 dB and 46.6 dB, respectively. These values include splices, FIFO insertion losses (0.35 dB average), and loss from the WDM couplers. The link losses were compensated by two high-power EDFA boosters (with same gain spectra and 31 dBm output) and two sets of backward-pumped DRAs using three semiconductor laser diodes at 1429 nm, 1447 nm, and 1465 nm.

Transmission Results and Discussion

The 800Gb/s DWDM transmission performance of the 2-core MCF unrepeated system (unidirectional configuration) was optimized by adjusting the launch signal power and the backward Raman pump powers. This optimization aimed to minimize real-time BER by obtaining high received OSNR while mitigating nonlinear impairments and maintaining C-band gain flatness. We applied signal pre-emphasis to the transmitter to counteract the inherent tilt in the Raman gain profile and the wavelength-dependent fibre loss, improving the Q^2 values for shorter wavelength channels. The optimized total launch power was approximately 27.2 dBm per core, with backward Raman pump powers of 1.40 W (Core 1) and 1.37 W (Core 2), providing an on-off Raman gain of 20~20.5 dB

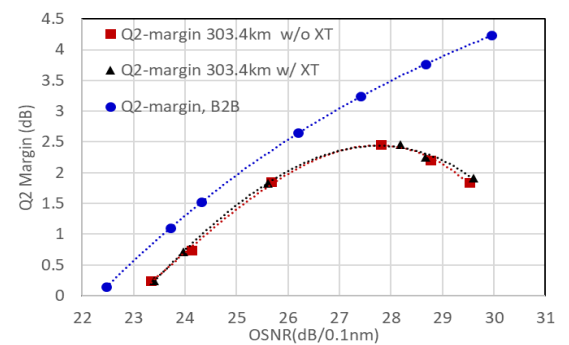


Fig. 3: Q^2 margin vs. OSNR for Core#1 at 1551.32nm, comparing for two cases: with XT and without XT.

A primary concern in MCF systems is the potential for inter-core crosstalk influence, particularly in high-baud-rate systems using sensitive modulation formats like PCS-16QAM, we first evaluated the impact of inter-core XT on an 800 Gb/s signal at 1551.32 nm in Core 1. Fig. 3 displays the Q^2 margin versus linear OSNR after 303.4 km. The performance was compared for two cases: with XT (signals transmitted unidirectionally in both cores) and without XT (signals only transmitted in core 1, with the core 2 booster

EDFA disabled). As shown in Fig. 3, there was no measurable CO-XT penalty. The back-to-back (B2B) Q^2 margin was also plotted for reference.

Subsequently, we characterized all 33 channels across the C-band at the optimized operation condition. For each core, measurements were performed with and without XT. The real-time BER were recorded and the OSNR were measured when the channels are scanned by wavelength, Fig. 4(a) and (b) show the Q^2 margin as a function of wavelength for both cores. The measured OSNR remained virtually identical in both cases, with average values of 26.4 dB for Core 1 and 25.6 dB for Core 2 (0.1 nm noise bandwidth). For Core 1, the minimum and average Q^2 -margins were 0.49 dB and 1.50 dB, respectively. For Core 2, the minimum and average Q^2 margins were 0.45 dB and 1.72 dB.

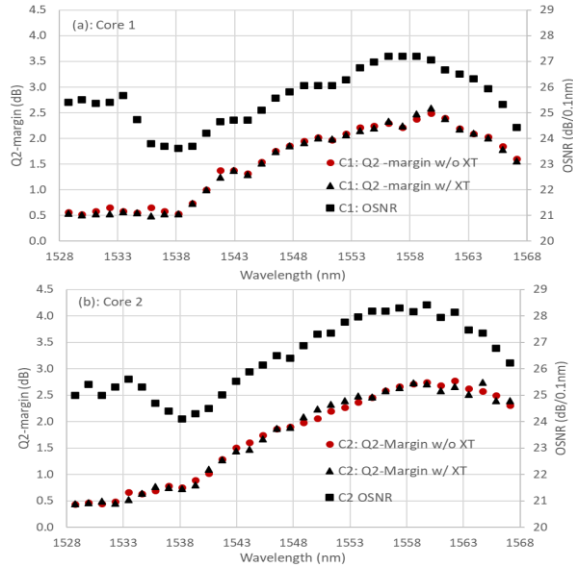


Fig. 4: Measured Q^2 margin and OSNR across full C-band (33 channels) for core 1 (a) and core 2 (b) after 303.4km unrepeated transmission.

Finally, the experimental setup was reconfigured to a bidirectional transmission. As shown in Fig. 1, the output of Booster 2 was connected to "FO" and launched into 303.4km 2-core MCF link at point B (red dashed line), while "FI" at point A was routed to the receiver (orange dashed line). All operating conditions remained unchanged from the unidirectional case. We scanned the channels by wavelength, recording BER and measuring OSNR. Fig. 5(a) and (b) display the Q^2 margin as a function of wavelength for both cores. The measured OSNR remained consistent with the unidirectional cases (not plotted). The minimum and average Q^2 -margins were 0.31 dB and 1.52 dB for core 1, and 0.25 dB and 1.57 dB for core 2, respectively. The minimum and average Q^2 margin for both bidirectional and unidirectional are summarized in table 3. The difference of average Q^2 margin between the "with XT" and

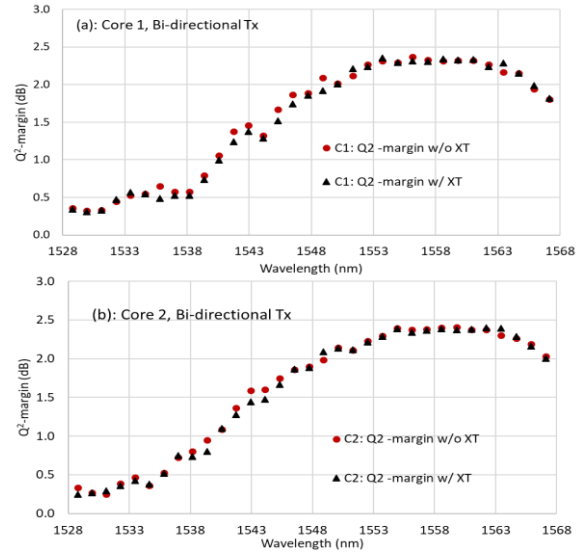


Fig. 5: Measured Q^2 margin across full C-band (33 channels) for core 1 (a) and core 2 (b) after 303.4km bi-directional unrepeated transmission.

Table 2: Measured CO-XT and Counter-XT from difference of 2-core MCF length (average)

		Unidir w/o XT	Unidir w/XT	Bidir w/o XT	Bidir w/ XT
Average Q^2 -margin (dB)	Core 1	1.52	1.50	1.54	1.52
	Core 2	1.72	1.72	1.59	1.57
Minimum Q^2 -margin (dB)	Core 1	0.52	0.49	0.32	0.31
	Core 2	0.43	0.45	0.25	0.25

"without XT" cases was less than 0.02 dB in both unidirectional and bidirectional configurations, confirming that inter-core crosstalk induces no measurable penalty.

Conclusions

We have successfully demonstrated a real-time unrepeated transmission of 52.8 Tb/s over a 303.4 km uncoupled 2-core MCF link. This record result was achieved using commercial high-speed pluggable modules and SCUBA 2X 2-core MCF, which features an ultra-low average attenuation of 0.147 dB/km and ultra-low CO-XT with an average value of -59 dB/100km. Our results confirm that even in a high-power transmission configuration using backward-pumped DRA, the inter-core crosstalk induces no measurable penalty in either unidirectional or bidirectional configurations. These findings validate the readiness of SCUBA 2X fiber for high-capacity terrestrial and subsea unrepeated branch applications.

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References

- [1] T. Matsui, Y. Sagae, T. Sakamoto, and K. Nakajima, "Design and Applicability of Multi-Core Fibers with Standard Cladding Diameter", *Journal of Lightwave Technology*, vol. 38, no. 21, pp. 6065–6070, 2020.
- [2] D. Soma, et.al., "18,090-km 2-core Fiber Transmission Using Circulatory Directional Fan-in/Fan-out Devices" OFC2024 Th4A.4
- [3] K. Komatsu, et.al., "126-Tb/s 2-Core Fibre Transmission over 114.6 km $\times$ 5 Spans with Ultra-Low-Loss Fibre Bundle Fan-in/Fan-out" ECOC2024, W3B.5,
- [4] T. Hayashi, T. Nagashima, A. Inoue, H. Sakuma, T. Suganuma, and T. Hasegawa, "Uncoupled Multi-core Fibre Design for Practical Bidirectional Optical Communications", M1E.1, OFC 2022.
- [5] T. Ito, E. L. T. de Gabory, M. Arikawa, Y. Hashimoto and K. Fukuchi., "Reduction of influence of inter-core crosstalk in MCF with bidirectional assignment between neighboring cores", OTh3K.2, OFC 2013.
- [6] H. Bissessur, A. Busson, F. Hedaraly, and D. Kravchenko, "Real-time 25.2 Tb/s/core unrepeated transmission over 256 km of 2-uncoupled core MCF Fibre", M1H.2, OFC 2025
- [7] H. Bissessur, A. Busson, F. Hedaraly, "Influernce of inter-core crosstalk in high -capacity 205 to 359km unrepeated transmission over 2-core MCF" M.03.05.5 ECOC2025.
- [8] B. Zhu et al., "25.6 Tbit/s (64x400Gb/s) Real-time Unrepeated Transmission over 320 km Fibres by 400ZR+ Modules", We3C1.4, ECOC2021
- [9] S. Almonacil et al, "C-band unrepeated transmission System over 291-km at net throughput of 41.3-Tbps without ROPA", ThB2.3, ECOC 2023
- [10] I. de Jauregui Ruiz et al, "66.8 Tb/s Real-Time C+L Unrepeated Transmission over 301 km using Forward and Backward Raman Amplification", W3F.4, OFC 2024.
- [11] R. S. Luis et al, "372 Tb/s Unrepeated 213 km Transmission Over a 125 μ m Cladding Diameter, 4-Core MCF", M4B.3, OFC2022
- [12] Stegmaier, M. et al. "2-CORE MULTICORE FIBER FOR SUBMARINE CABLE SYSTEMS WITH ULTRA-LOW AVERAGE LOSS OF 0.147 DB/KM", We1A-1 Suboptics 2025
- [13] D. Kravchenko, H. Bissessur, F. Hedaraly and A. Busson, "Crosstalk Requirement for Multi-Core Optical Fibres in Unrepeated Systems with Distributed Raman Amplification", MoB1.1, OECC 2024.